

Grace Fanson

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Earth Atmospheric & Planetary Sciences | Massachusetts Institute of Technology

RESEARCH INTERESTS

I study planetary surface evolution with the goal of understanding how planetary bodies have developed into their present-day states. My work integrates a range of research methods including numerical modeling, remote sensing data analysis, and field studies at terrestrial analog sites to explore surface processes. I am particularly interested in developing our understanding of the early water cycle on Mars.

EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, Massachusetts

Doctor of Philosophy (PhD) in Planetary Science

Est. 2028

- Department | Earth, Atmospheric & Planetary Sciences
- Advisor | Dr. Gaia Stucky de Quay

Georgia Institute of Technology

Atlanta, Georgia

Bachelor of Science (BS) in Physics

2018 - 2022

- School of Sciences | Physics
- Advisors | Dr. James Wray & Dr. Frances Rivera-Hernández

RESEARCH EXPERIENCE

Gaia Lab

2023 - present

- Institution | Massachusetts Institute of Technology
- Advisor | Dr. Gaia Stucky de Quay
- Project Title | Quantifying valley network preservation on Mars after impact cratering.

Project Description

- Impact cratering model created in Python to simulate the destruction of valley networks on Mars.
- Synthetic landscape developed with the Python toolkit LandLab used as starting topography
- Principles of Geomorphology applied to eroded valley volume to calculate volume of valley carving precipitation

MIT Glaciology

2025 - present

- Institution | Massachusetts Institute of Technology
- Advisor | Dr. Brent Minchew
- Project Title | Quantifying the effects of surface stretching on the morphology of craters on Enceladus

Project Description

- Developed an ice stretching model representing the global ice shell of Enceladus in Python based on the fundamentals of fluid mechanics

- Institution | University of Cambridge | Trinity College
- Advisor | Dr. Emily Bamber
- Project Title | Modeling the relationship between rainfall and river valley depth as a proxy for Martian valley incision

Project Description

- Developed landscape evolution model in the python toolkit LandLab to simulate valley incision to build relationship between cumulative precipitation and valley incision depth

ARES Lab / PLANETAS Group

2021 - 2023

- Institution | Georgia Institute of Technology
- Advisors | Dr. James Wray and Dr. Frances Rivera-Hernandez
- Project Title | Assessing the role of ancient volcanism in northern Arabia Terra, Mars

Project Description

- Analyzed CRISM data using ENVI software to determine if mineralogy in Northern Arabia Terra is consistent with past volcanic activity
- Developed Python tool to aid in analysis of silica spectra

FELDSPAR

2022 - 2023

- Institution | Georgia Institute of Technology & University of Idaho
- Advisor | Dr. Erika Rader & Dr. Anna Simpson
- Project Title | Large-scale sampling of biosignature distribution in aeolian- and volcanism-resurfaced Icelandic Mars analogs: integrating ground data, artificial landscape models, and satellite data

Project Description

- Analyzed WorldView-3 data using QGIS Software to investigate potential biosignatures at Mars analogue sites in Iceland

PUBLICATIONS

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- Hughes, E., Wray, J., Karunatillake, S., **Fanson, G.**, Harrington, E., Hood, D. (2024) Water-Limited Hydrothermalism and Volcanic Resurfacing of Eridania Basin, Mars. JGR-Planets. DOI: 10.1029/2024JE008461

In Revision

- **Fanson, G.**, & Stucky de Quay, G. Quantifying valley network preservation on Mars after impact cratering. npj Space Exploration (in prep, Est. Sept 2026)
- **Fanson, G.**, Wray, J., Rivera-Hernández, F., Hughes, E. Assessing the role of ancient volcanism in northern Arabia Terra, Mars. JGR-Planets (in prep, Est. 2025)

PRESENTATIONS & TALKS

Conference Presentations

* = oral

- ***Fanson, G.**, Morgan, A., & Stucky de Quay G. (2026) Quantifying the effects of impact cratering on Martian valley network preservation. LPSC
- **Fanson, G.**, & Stucky de Quay, G. (2024). Impact events destroy Martian valley networks. AGU Fall Meeting
- **Fanson, G.**, & Stucky de Quay, G. (2024). Constraining the fraction of ancient Martian valleys destroyed by impact events. Tenth International Conference on Mars
- ***Fanson, G.** & Stucky de Quay. (2024). Constraining the Fraction of Ancient Martian Valleys Destroyed by Impact Events. Boston Area Planetary Science Meeting.
- ***Fanson, G.**, Wray, J., Rivera-Hernández, F., Hughes, E., & Matiella Novak, A. (2022). Further evidence for ancient explosive volcanism in Northern Arabia Terra, Mars. GSA Annual Meeting.
- Simpson, A., Rader, E., Pital, A., Zinzer, B., **Fanson, G.**, Wroblewski, F., Wray, J., Cable, M., Stockton, A., & FELDSPAR. (2022). Large-scale sampling of biosignature distribution in aeolian- and volcanism-resurfaced Icelandic Mars analogs: integrating ground data, artificial landscape models, and satellite data. AbSciCon22
- ***Fanson, G.** & Wray, J. (2022). Testing the hypothesis of ancient volcanism in Arabia Terra. ExplOrigins Colloquium

Invited Talks

- **Imperial College London** | Department of Earth Science and Engineering 2025
- **University of Cambridge** | Department of Earth Sciences 2025

GRANTS & FELLOWSHIPS

Massachusetts Institute of Technology

- NSF Graduate Research Fellowship Program (full tuition + stipend) 2025 - 2028
- MIT Office of Graduate Education Conference Grant (\$500) 2026
- Massachusetts Space Grant Fellowship (\$7,600) 2025
- Professor Patrick M. Hurley Graduate Fellowship (full tuition + stipend) 2024 - 2025
- Travel Award - 10th International Conference on Mars 2024
- Praecis Presidential Graduate Fellowship (full tuition + stipend) 2023 - 2024

Georgia Institute of Technology

- Travel Award - Department of Earth and Atmospheric Science (\$1,000) 2022
- President's Undergraduate Research Award (\$1,500) 2022
- College of Sciences Dean's Scholarship (in-state tuition ~\$80,000) 2018 - 2022

WORKSHOPS

- Social Field Safety - Field Inclusive 2026

ACADEMIC AWARDS / HONORS

Georgia Institute of Technology

- Highest Honors in Physics 2022
- Dean's List 2018 - 2022

SERVICE AND OUTREACH

Application Mentorship Program (AMP) at MIT 2024 - present

- Organize AMP events and coordinate mentorship
- Mentor to applicants to EAPS graduate programs

Volunteer at Cambridge Science Festival 2024

- Worked as a part of the "Ask a Scientist Booth"

Cambridge School Volunteers 2023 - 2024

- Tutor 6th grade student in math and science

Atlanta Science Festival 2022

- Created and led spectroscopy and impact cratering workshops for children

Science and Entertainment Exchange 2021

- Moderated and organized live panel discussion

MENTORING EXPERIENCE

Noella Aldrich 2026

Undergraduate Research Opportunity Program (UROP) at MIT

- Institution | Massachusetts Institute of Technology
- Project Title | Quantifying the alteration of Martian valley drainage density post impact

Charlotte Whitworth 2024 - 2025

Undergraduate Research Opportunity Program (UROP) at MIT

- Institution | Imperial College London
- Project Title | Studying Martian Volcanic River Valleys

PROFESSIONAL EXPERIENCE

Intro Geophysics & Planetary Science Teaching Assistant (MIT) 2024

- Led class discussions

Optics Teaching Assistant (Georgia Tech) 2022

- Held office hours and graded assignments / exams

Systems Engineering Intern at Sierra Nevada Corporation Space Systems Group 2020

- Developed requirements for Crew, Internal Thermal Control, Environmental Control, and Life Support systems for the Human Landing System
- Aided development of test plans for the Thermal Control System for Dream Chaser

TECHNICAL SKILLS

Software

- **Mapping Software** | ArcPro, QGIS, etc.
- **Python** | Arcpy, GDAL, Landlab
- **Microsoft Office** | Word, Excel, Power Point, etc.
- **ENVI/IDL** | CRISM Analysis Toolkit (CAT)
- **Adobe Illustrator** | Figures & Schematics
- **L^AT_EX**

Field Work

Catalina Island 2024

- Deploying seismometers
- GPS measurements
- Calibrating and taking gravity measurements

Baker River New Hampshire 2023

- Measure river valley profile
- Grain Size measurements
- Water flow velocities

Data Sets

- Compact Reconnaissance Imaging Spectrometer (CRISM)
- High Resolution Imaging Experiment (HiRISE)
- WorldView III
- Context Camera (CTX)

- Mars Orbiter Laser Altimeter (MOLA)
- Thermal Emission Imaging Spectrometer (THEMIS)
- Thermal Emission Spectrometer (TES)